

The circuit can produce one ampere with one volt in a sample the diameter of a human hair and 2.5cm long. However, only about 10 per cent of the photovoltaic circuits on a glass side work, which if increased can increase the amount of charge being produced as well.

These photovoltaic circuits can also be ideal for powering tiny robotic devices or running computer calculations at the speed of light.

Flexible solar cells made with silicon wires

Using arrays of long, thin silicon wires embedded in a polymer substrate, a team of scientists from the California Institute of Technology (Caltech) has created a new type of flexible solar cell that enhances the absorption of sunlight and efficiently converts its photons into electrons. The solar cell does all this using only a fraction of the expensive semiconductor materials required by conventional solar cells.

The light-trapping limit of a material refers to the amount of sunlight that it is able to absorb. The silicon-wire arrays absorb up to 96 per cent of the incident sunlight at a single wavelength and 85 per cent of the total collectible sunlight.

The silicon wire arrays created by the Caltech team can convert between 90 and 100 per cent of the photons they absorb into electrons—in techni-

cal terms, the wires have near-perfect internal quantum efficiency.

Flexible material that captures energy

Researchers at Princeton University have developed a material that when stressed records energy. This material contains ribbons of a piezoelectric material called PZT between pieces of silicone. When it is manually stressed, the piezoelectric material generates a voltage that can be used to produce electric current. The current generated can later be converted back into me-

chanical movement.

According to Princeton University researchers, this material can be incorporated into the soles of shoes to power portable electronics or even placed in a patient's lungs to recharge the pacemaker as he breathes.

New way to build fuel-cell membranes

Researchers at MIT and Pennsylvania State University are developing a level-by-level assembly method for producing novel kinds of membranes that could have improved properties

for batteries, fuel cells and other energy conversion and storage applications. They have come up with a material specifically designed for advanced direct-methanol fuel cells—devices that can convert fuel into electricity without combustion, thereby avoiding the emission of any pollutant or greenhouse gas.

Electrolytes, used in both batteries and fuel cells, are materials that contain many ions (atoms or molecules that have a net electrical charge), making it easy for an electric current to flow through them. In both batteries and fuel cells, this material is sandwiched



A piece of silicone rubber imprinted with super-thin material that generates electricity when flexed. The technology could provide a source of power for mobile and medical devices. (Credit: Frank Wojciechowski)

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